MONASH University

# EXPLANATORY STATEMENT

**Project ID: 51583**
**Project title: Teaching neural network fitting**


| **Chief Investigator:** Prof Dianne Cook<br>Department of Econometrics and Business Statistics, Monash University<br>Phone: +61 03 990 52608<br>email: dicook@monash.edu | **Collaborators:**<br>Janith Wanniarachchi,<br>Janith.Wanniarachchi@monash.edu<br>Dr Jack Jewson<br>Jack.Jewson@monash.edu |
|---|---|


During the workshop, you'll be asked a few questions to help explore the ideas we're covering. By providing a response, you are agreeing to allow us to use your data. **You can choose not to respond to any question**. Your answers are anonymous, cannot be individually identified, and only summary statistics are stored. Nothing will be marked or affect your grade. You will be able to use a web app during the session, which does not collect any data.

Participating in these activities can *help you build your understanding,* and providing your responses will help us *improve how we teach this topic*.

**What does the research involve?**

This study investigates whether the use of a new web app in classroom workshops can improve students' understanding of model complexity and enhance their model-fitting skills. Students attending the regular weekly workshops, either in person or remotely, will receive additional instructional content on model fitting through lecture slides and guided demonstrations of the app. Before and after the session, students will be asked to respond to questions via Mentimeter to assess changes in understanding. Participation is voluntary, and students can opt out of any activity or question without penalty. Responses are collected anonymously, and only aggregated summary data will be recorded.

The research student, who developed the web app, will conduct the instructional sessions and collect aggregated summary data. They have several years of experience teaching statistics and data science, ensuring that the sessions are delivered effectively. While they have not previously conducted formal learning experiments, they will follow a structured study protocol and are supervised by experienced researchers to ensure ethical and rigorous data collection.

The chief examiner will attend the sessions only as an observer and will not access any participant data. No personally identifiable information will be collected at any stage, and all aggregated data will be stored securely on the researcher's password-protected computer during the study. For purposes of publication and transparency, aggregated summary data may be shared on a secure GitHub repository, but no individual responses or identifying information will be included.

**Why were you invited for this research?**

You are invited to participate in this research because you are enrolled in either ETC3250 or ETC5250 in semester 1, 2026 at Monash University.

**Opt out of the project**

You can opt out by simply **choosing not to submit responses** to any Mentimeter questions. You can also skip any individual question if you do not wish to answer. Opting out has no impact on your participation in the class, grades, or standing in the course. This process ensures that you retain full control over their participation while maintaining the anonymity and integrity of the study.

**Possible benefits and risks to participants**

This research involves minimal risk to participants, consistent with normal classroom activities, while providing potential benefits such as improved learning and insights into teaching practices that may benefit future students and the broader educational community.

**Payment**

Participants will not receive any payment or other incentives for taking part in this study. Participation is entirely voluntary, and students may choose to opt out of any activity without penalty. As the research activities are embedded within regular classroom sessions and involve minimal risk, no financial or material incentive is required.

**Confidentiality**

All data collected in this study will be treated as strictly confidential. No personally identifiable information will be recorded at any stage, and all responses will be anonymous. Aggregated summary data from Mentimeter will be manually recorded and stored securely on the researcher's password-protected computer. Only the research team will have access to these data.

Any data used for analysis or publication will also be aggregated and non-identifiable, ensuring that individual participants cannot be recognised. These measures maintain participant confidentiality while allowing the research to achieve its educational and scientific objectives.

**Storage of data**

During the study, aggregated summary data will be manually recorded and stored on the researcher's password-protected computer. No individual responses or identifying information will be stored. Access to this computer will be limited to the research team, and data files will be stored in secure folders in accordance with university data security policies.

After analysis, only aggregated, non-identifiable data used for publications will be uploaded to a GitHub repository to support transparency and reproducibility. This approach ensures that participant privacy is maintained while enabling secure management of study data and responsible sharing of results.

**Use of data for other purposes**

In accordance with data sharing guidelines, de-identified data may be made available for use by other researchers. This data will be held on public repository(s) as may be a requirement of journals for publication. Shared data will not include any personally identifying details.

**Results**

The results will be shared at the end of the workshop. You can also access the results by emailing any of the researchers, at the addresses provided above.

**Complaints**

Should you have any concerns or complaints about the conduct of the project, you are welcome to contact the Executive Officer, Monash University Human Research Ethics Committee (MUHREC):

| | |
|---|---|
| Executive Officer<br>Monash University Human Research Ethics Committee (MUHREC)<br>Office of Research Ethics and Integrity<br>Room 116, Administration Building B (3D)<br>26 Sports Walk, Clayton Campus<br>Monash University VIC 3800<br><br>Tel: +61 3 9905 2052 Email: muhrec@monash.edu | |